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Recursion ✓

↓

DP me swaad aa jayega

# Dynamic Programming

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# Checklist

1. Memorization & Tabulation
2. 1D DP Problems
3. 2D DP Grid Problems
4. Knapsack & Unbounded Knapsack Problems
5. DP on Strings
6. MCM problems and DP on Trees

# What is Dynamic Programming?

maths ka topic - desh

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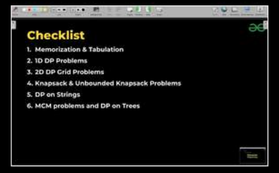
Programming

Optimised recursion

overlapping subproblems



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# Problem

## Nth Fibonacci Number (Recursion)

```
int fibo(n){
    if(n==0 || n==1) return n;
    return fibo(n-1) + fibo(n-2);
}
```

$T.C. = O(2^n)$

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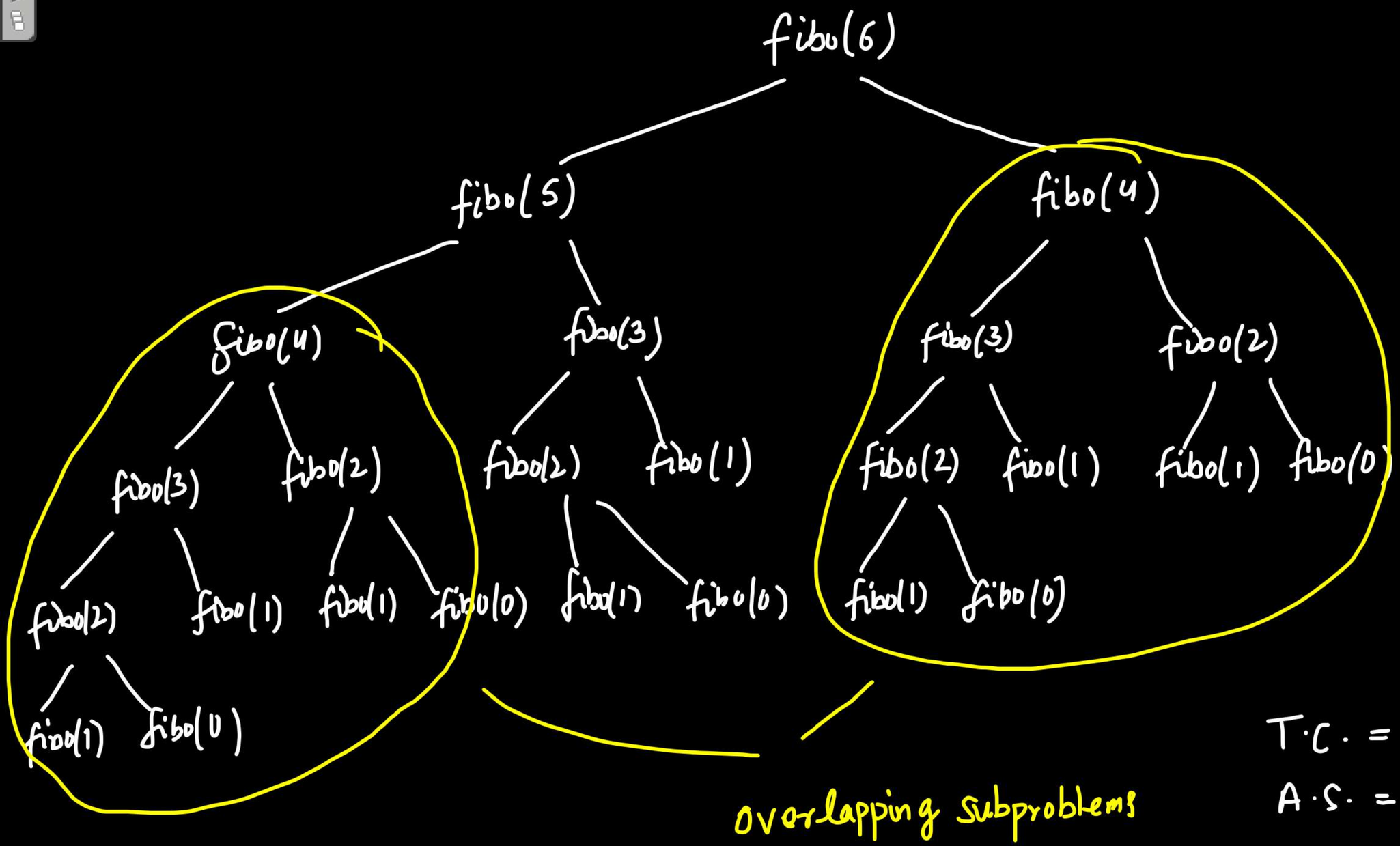
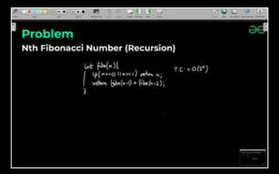



$T.C. = O(2^n)$

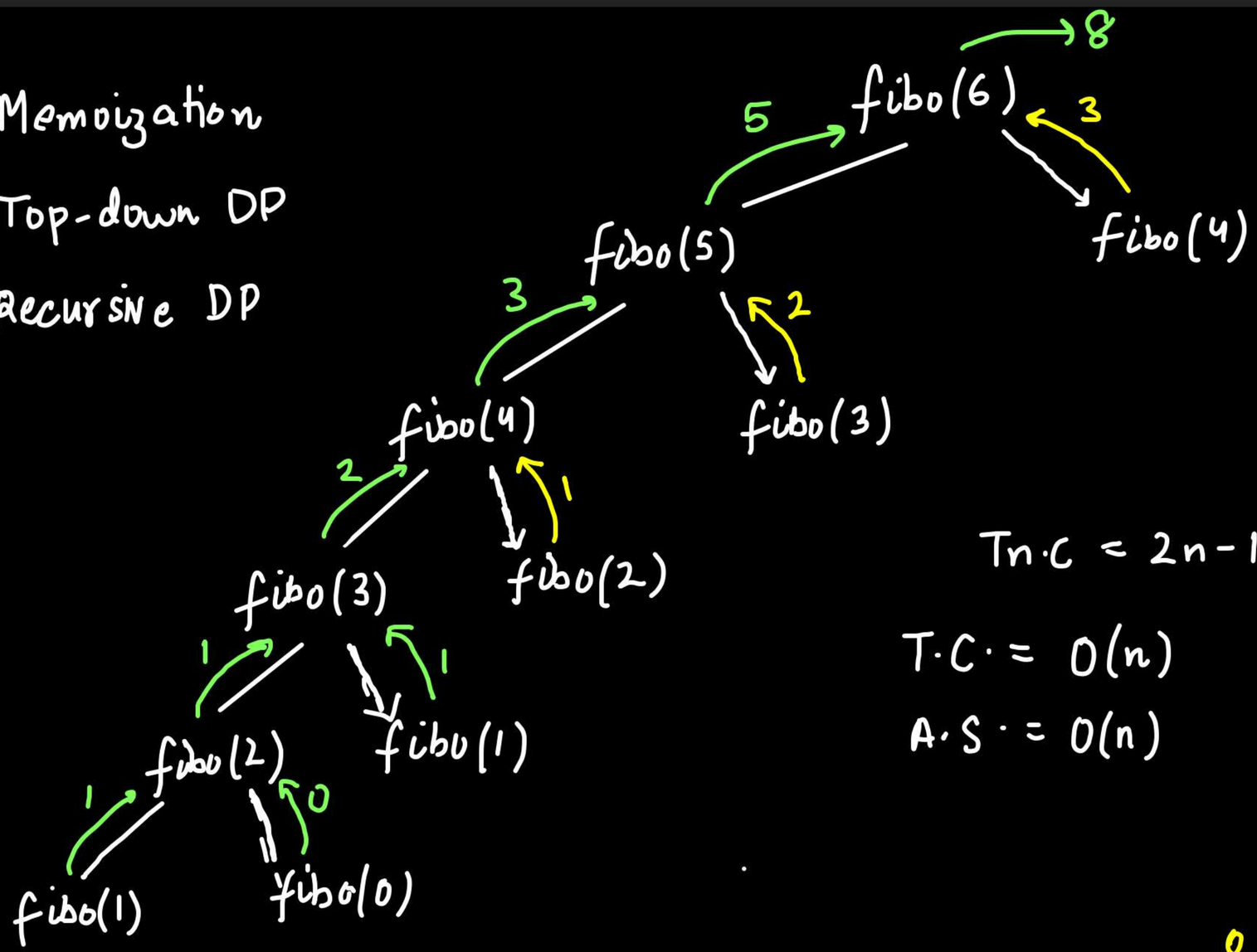
$A.S. = O(n)$



Memoization

Top-down DP

Recursive DP



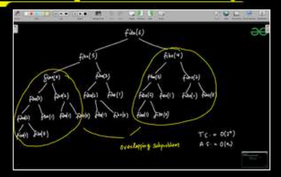

```
static int[] dp;
public int fibo(int n) {
    if(n<=1) return n;
    if(dp[n]!=0) return dp[n]; // extra
    int ans = fibo(n-1) + fibo(n-2);
    dp[n] = ans; // extra
    return ans;
}
public int nthFibonacci(int n) {
    dp = new int[n+1]; // idx from 0 to n
    return fibo(n);
}
```

$T_n \cdot C = 2n-1$

$T \cdot C \cdot = O(n)$

$A \cdot S \cdot = O(n)$

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 |
|---|---|---|---|---|---|---|---|
| dp | 0 | 0 | 1 | 0 | 0 | 0 | 0 |



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## Factorial of a Number

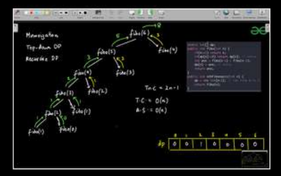
$$fact(n) = n * fact(n-1)$$

5
↓
4
↓
3
↓
2
↓
1

## Power

$$pow(a,b) = a * pow(a,b-1)$$

When $f^n$ calls itself only once, Then DP makes no difference.



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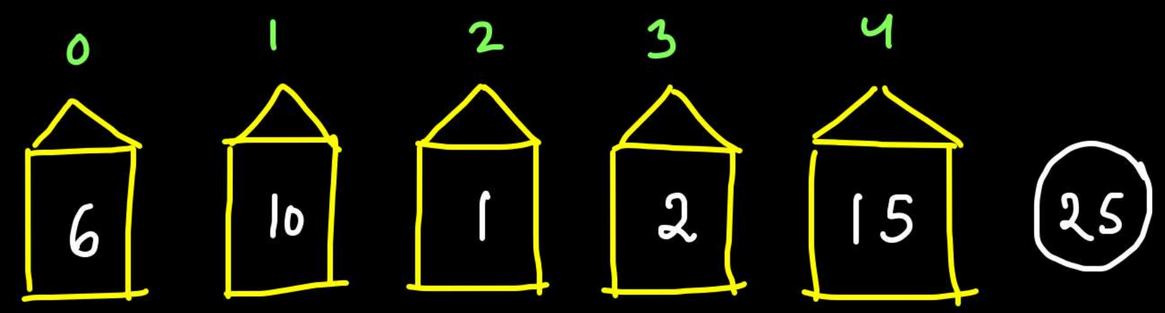
# Problem

## Stickler Thief

2 adjacent gharo se chori not allowed

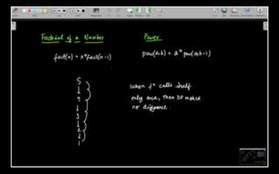
vivek




i pick or skip

$\downarrow$ $\downarrow$

$i+2$ $i+1$



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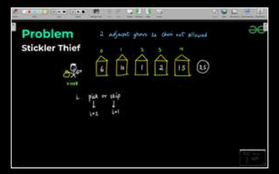
## Homework :

1) $n^{th}$ stair

2) House Robber 2 (LC-213)



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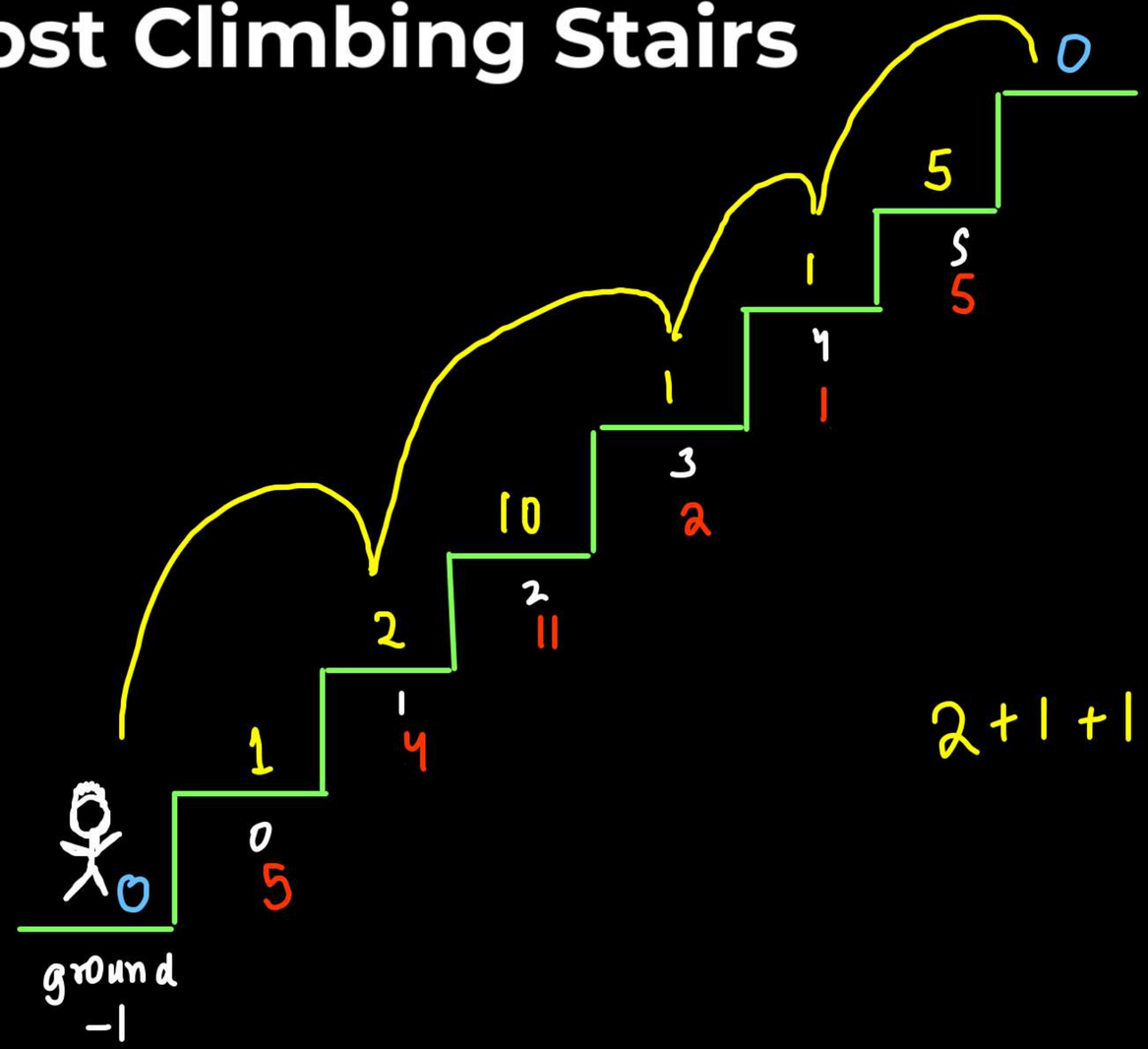
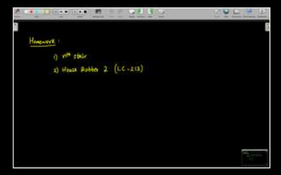
# Problem

## Min Cost Climbing Stairs




bishal
↓
1 or 2
jump

$2+1+1=4$